

PAPER NO. 2104 - SELECTED QUESTIONS

PERMUTATIONS | COMBINATIONS | ALGORITHMS | RECURRENCE RELATIONS

Question numbers follow the supplied source questions.

PERMUTATIONS AND COMBINATIONS

5. In how many ways can we select a chairperson, vice-chairperson, secretary, and treasurer from a group of 10 persons?
6. In how many ways can we seat seven men and five women in line if no two women stand together?
14. How many arrangements of the letters of the word WOMEN can be made if every arrangement is to begin and end with the vowels?

ALGORITHM - NEXT PERMUTATION

3. Find the permutation that will be generated by Algorithm after the permutation 12354.
4. Find the permutation that will be generated by Algorithm after the permutation 625431.
5. Find the permutation that will be generated by Algorithm after the permutation 12876543.

RECURRENCE RELATIONS - COMPOUND INTEREST

- A. A person invests 2000 K at 14 percent compounded annually. Let A_n represent the amount of the investment at the end of n years.
 - (i) Find a recurrence relation for the sequence $A_0, A_1, \dots$
 - (ii) Find an initial condition for the sequence $A_0, A_1, \dots$
 - (iii) Find A_1, A_2 and A_3 .
 - (iv) Find an explicit formula for A_n .
 - (v) How long will it take for a person to double the initial investment?
- B. A person invests 2000 K each year in a tax-sheltered annuity at 10 percent compounded annually. Let A_n represent the amount at the end of n years.
 - (i) Find a recurrence relation for the sequence $A_0, A_1, \dots$
 - (ii) Find an initial condition for the sequence $A_0, A_1, \dots$
 - (iii) Find A_1, A_2 and A_3 .
 - (iv) Find an explicit formula for A_n .
- C. A person invests 3000 K at 12 percent annual interest compounded quarterly. Let A_n represent the amount at the end of n years.
 - (i) Find a recurrence relation for the sequence $A_0, A_1, \dots$
 - (ii) Find an initial condition for the sequence $A_0, A_1, \dots$
 - (iii) Find A_1, A_2 and A_3 .
 - (iv) Find an explicit formula for A_n .
 - (v) How long will it take for a person to double the initial investment?